Measuring $^{12}CO/^{13}CO$ in JWST Transmission Spectra: A Case Study with WASP-39b

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Introduction

Goal:

Can we measure $^{12}CO/^{13}CO$ in the JWST transmission spectrum of WASP-39b, gaining insights into hot Jupiter formation?

Context:

- $^{12}CO/^{13}CO$ can shed light on formation history
- Not yet measured from a JWST transmission spectrum
- Hints of ^{13}CO in WASP-39b[1,2]

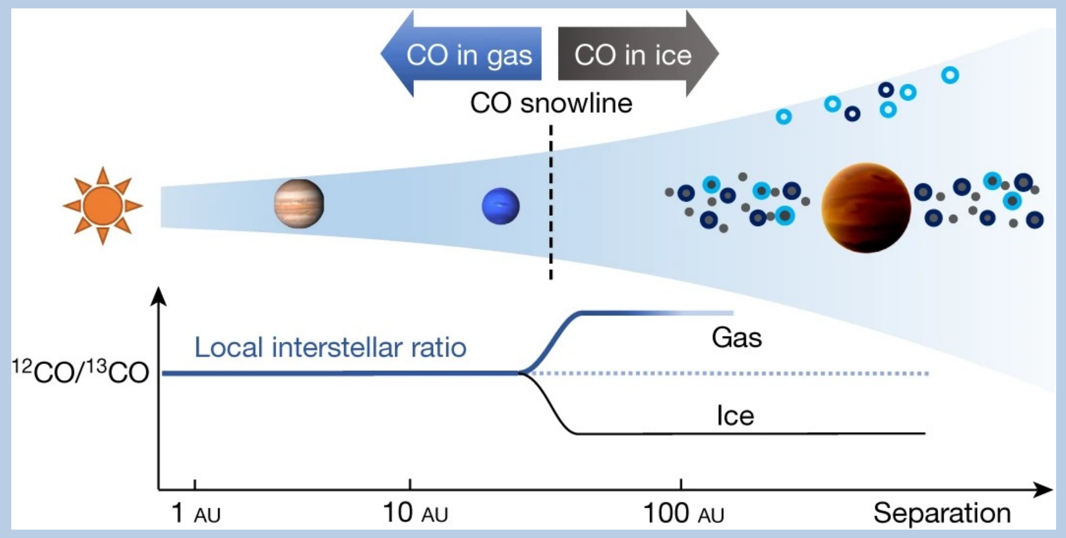
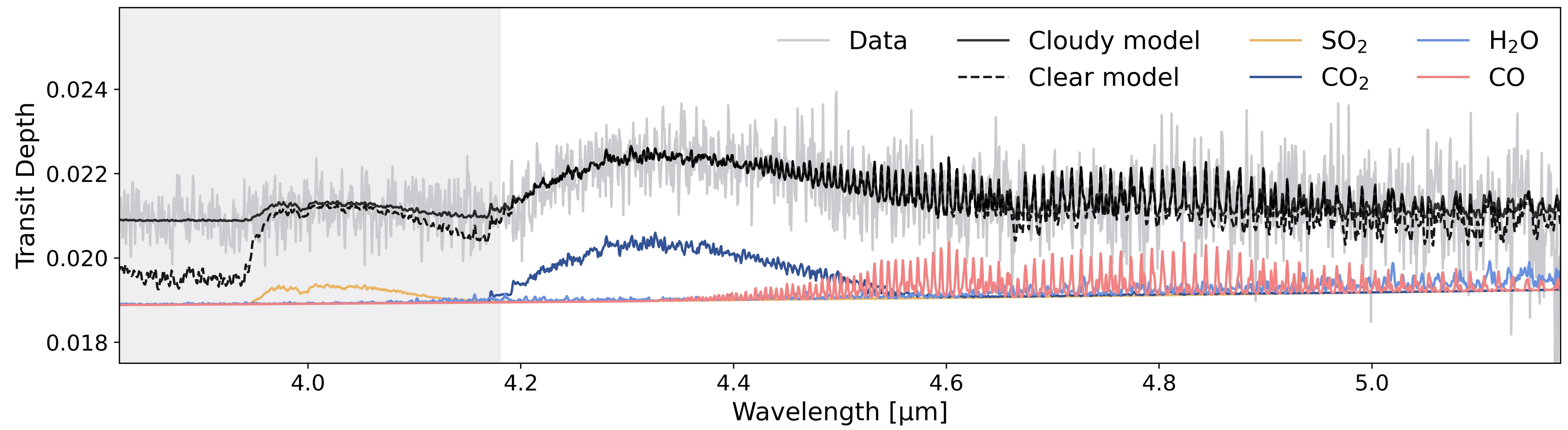


Figure 1: Icy regions beyond CO snowline might be enriched in ^{13}CO from carbon fractionation[3]

Findings:

- We detect ^{13}CO, but with a large 1σ uncertainty
- Noise biases our models to higher ^{13}CO abundances
- More observations and higher S/N data can help us measure this ratio more precisely

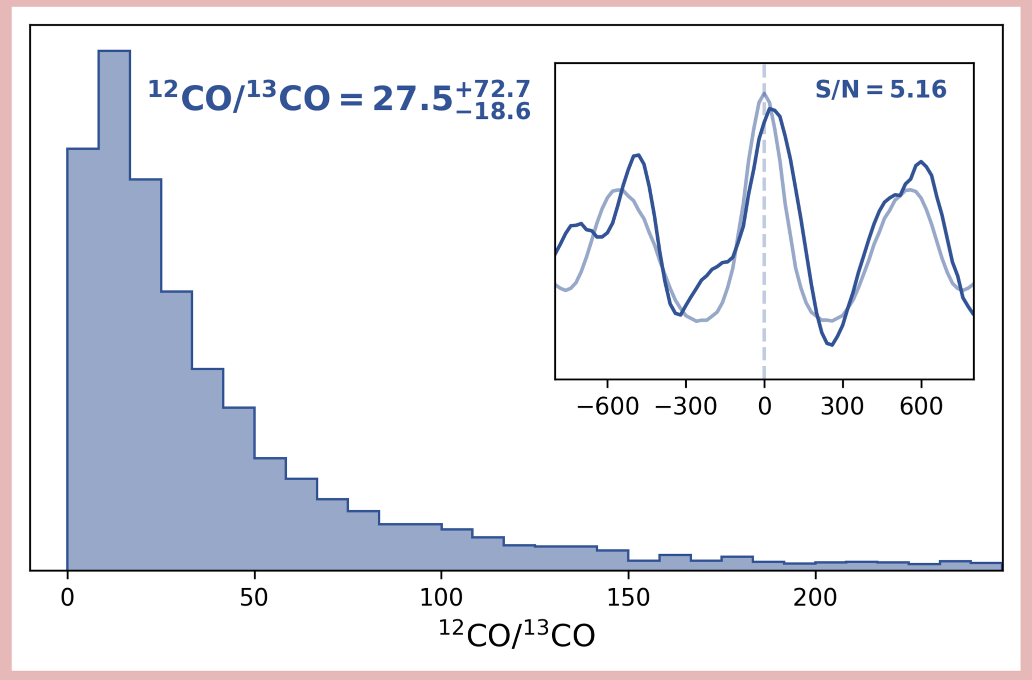


Methods

- JWST NIRSpec/G395H transmission spectrum spans 3.82–5.18 μm, includes H_2, He, CO_2, CO, H_2O, SO_2
- Fit for planet radius, temperature, metallicity, C/O ratio, $^{12}CO/^{13}CO$, and radial velocity[4,5]

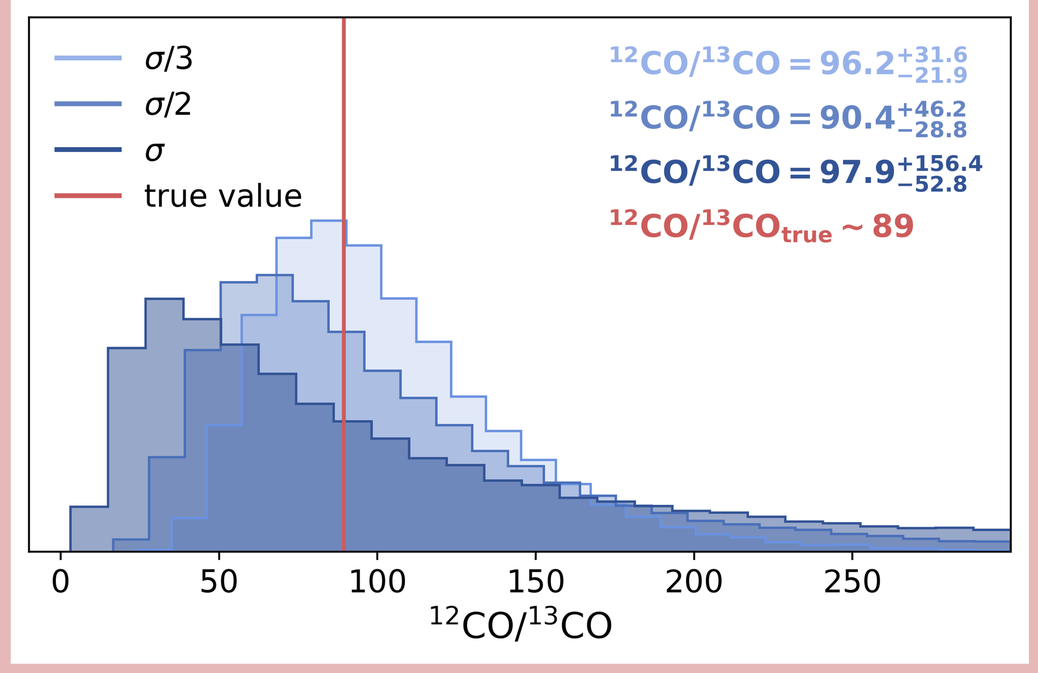
Results: WASP-39b

- Detect ^{13}CO with 5.2σ significance
- Low isotope ratio of $^{12}CO/^{13}CO$ ~ 28 (ISM ~ 66, solar ~ 89)
- Large 1σ uncertainty ($^{12}CO/^{13}CO$ ~ 8.9-100.2) due to lower signal-to-noise



Results: Simulated Data

- Bias to lower values at noise levels of the data (σ)
- ^{13}CO line width is inversely proportional to $^{12}CO/^{13}CO$
- Noise leads to Gaussian line strength distribution = skewed inverse distribution



Conclusions

- We measure a low isotope ratio (like WASP-77Ab[9])
- But error bars are large due to noisiness of data
- Noise biases retrievals to measure more ^{13}CO than is present
- More observations will enable more precise isotope measurements

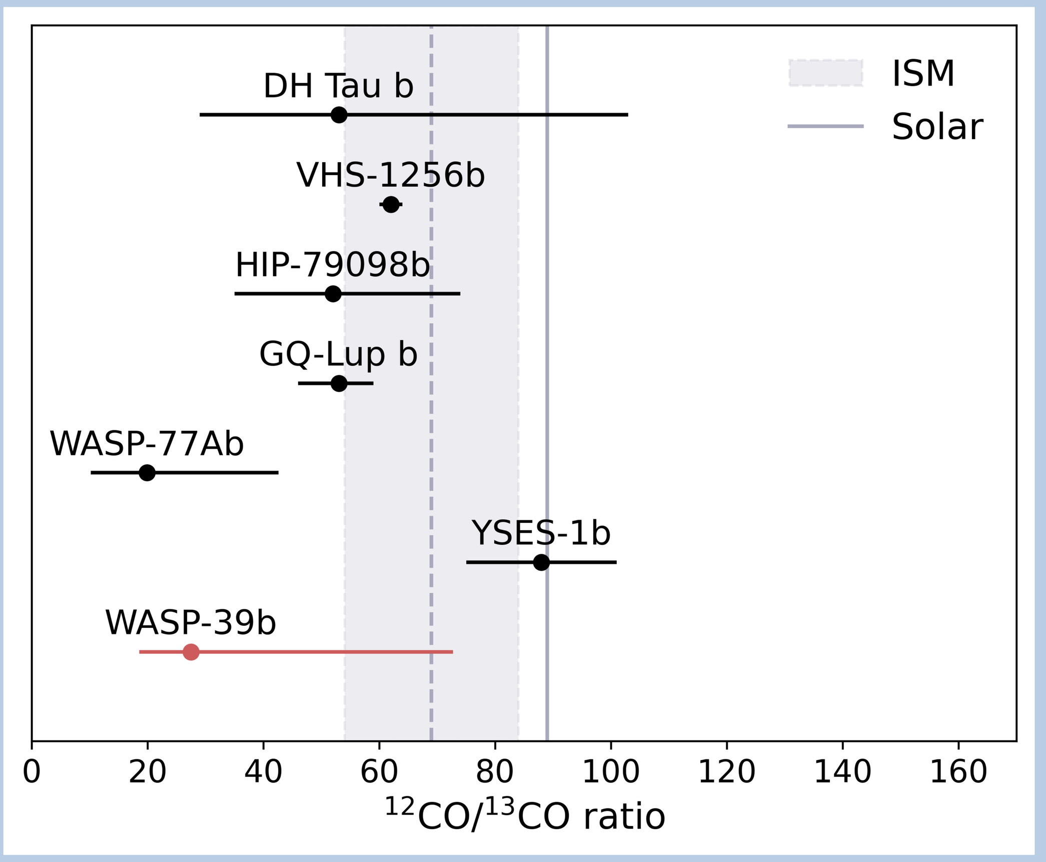


Figure 4: Comparing WASP-39b's isotope measurement to previous work[3,6,7,8,9,10,11,12]

References

[1]Faedi et al. 2011, [2]Esparza-Borges et al. 2023, [3]Zhang et al. 2021, [4]Mollière et al. 2019 (`petitRADTRANS`), [5]Buchner 2021 (`UltraNest`), [6]Xuan et al. 2024, [7]Gandhi et al. 2023, [8]González Picos et al. 2025, [9]Line et al. 2021, [10]Zhang et al. 2024, [11]Milam et al. 2005, [12]Clayton & Nittler 2004